

Partial moment momentum[☆]Yang Gao^a, Henry Leung^{b,*}, Stephen Satchell^c^a School of Economics, Huazhong University of Science and Technology, Wuhan 430074, China^b Business School, The University of Sydney, Sydney, NSW 2006, Australia^c Trinity College, University of Cambridge, Cambridge CB2 1TQ, United Kingdom

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ABSTRACT

While momentum benefits from persistent trends of the market, such strategies are unable to distinguish between upside and downside risk and suffer consequently. We propose a Partial Moment Momentum (PMM) trading strategy that is sensitive to the sign of risk and show risk-adjusted outperformance compared to plain momentum and volatility-adjusted momentum strategies. The outperformance is robust across multiple time periods and in particular during market downturns. Further analysis based on conventional linear factor models shows negligible exposure to factor risk for our PMM portfolio. Finally, the performance of our proposed strategy appears to be enhanced when time series momentum is present and allows for improved risk management by distinguishing between upside and downside risks.

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1. Introduction

Cross-sectional momentum strategies are implemented by buying previous winners and selling previous losers. It has been established that plain cross-sectional momentum strategies (henceforth, WML) are profitable in different markets and asset classes across different sample periods (e.g., Jegadeesh and Titman, 1993, 2001; Rouwenhorst, 1998; Chan et al., 2000; Okunev and White, 2003; Asness et al., 2013).

Recent literature has introduced Volatility-adjusted Momentum strategies (henceforth, VM) in both a cross-sectional momentum setting (e.g., Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016) and a time-series momentum setting (e.g., Moskowitz et al., 2012). For instance, Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016), show that gains from momentum strategies may be wiped out by extreme losses during crashes. Using the high level of predictability of the risk of momentum returns, the authors scale their long-short WML momentum portfolio and improve performance. Daniel and Moskowitz (2016) implement a dynamic momentum strategy based on conditional moments (mean and variance) and achieve

twice the alpha and Sharpe ratio compared to the WML portfolio over multiple time periods and different equity markets. This evidence suggests that VM strategies, which dynamically account for past volatility, act as a hedging mechanism for extreme momentum losses following sudden market downturns.¹ Moreira and Muir (2017) show that volatility-managed portfolios yield large alphas and increase Sharpe ratios in periods of high volatility. They also reveal significant risk-adjusted returns after controlling for equity risk factors, as well as Betting-Against-Beta (BAB) factors and a currency carry trade factor.² This work is extended by Barroso et al. (2020), who present a volatility-managed version of the BAB strategy. The authors show an improvement in the Sharpe ratio but they reveal that the strategy faces large downside risk compared to the risk-managed momentum strategy of Barroso and Santa-Clara (2015). The risk-managed BAB results are shown to be robust to the employment of volatility forecasts, the incorporation of leverage constraints and transaction costs and finally, exhibit consistency in results at the international level.

The empirical literature presents evidence that scaling the weights of momentum portfolios increases the Sharpe ratio of

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* Corresponding author.

E-mail addresses: gao_yang@hust.edu.cn (Y. Gao), henry.leung@sydney.edu.au (H. Leung), ses11@cam.ac.uk (S. Satchell).

¹ We construct a VM portfolio consistent with the risk-managed momentum strategy in Barroso and Santa-Clara (2015). The authors term their VM strategy the "risk-managed momentum" (Barroso and Santa-Clara (2015, p. 115)).

² The BAB strategy, proposed by Frazzini and Pedersen (2014), longs low-beta stocks and shorts high-beta stocks. They show that the Sharpe ratio for the strategy approximately doubles that of the U.S. equity market.

WML. However, it is well-known that volatility is larger when markets fall than when they rise. Furthermore, the economic consequences of volatility in falling or rising markets are not equal. Rather than scaling WML portfolios, which do not distinguish between upside and downside risk, we construct partial moment momentum strategies based on the partial moment decompositions of past squared market returns to better capture market trends and reduce losses following market turbulence. Our key contribution is to present results that reveal the outperformance of an unconstrained partial moment momentum strategy (PMM), which employs unequal scaling of positions between the long-side and short-side weights according to the relative levels of positive and negative semi-variances to distinguish itself from the VM strategy, which assumes equal positions on long-side and short-side weights.

We find that the annualized Sharpe ratio of our PMM strategy significantly outperforms the benchmark WML and VM strategies by 0.70 and 0.20, respectively, over the whole sample period from 1927 to 2018. In addition, the skewness of the PMM strategy improves from -0.23 to 0.03 compared to the VM portfolio over the same period. Compared to the VM strategy, the flexibility of the unequal long-short weight scaling element intrinsic to the PMM strategy is characterized by outperformance during market crashes through the management of downside risk. For instance, during the high volatility GFC period, the outperformance of our PMM strategy against the VM strategy is mainly derived from the tilting of the PMM portfolio scaling towards the short-side weight relative to the long-side weight. In the low volatility period prior to the GFC, the dynamic leverage of the PMM strategy demonstrates a proportional increase in short position dominated by a sharp decrease in the weight of the long leg compared to the short leg. During periods of contraction within the whole sample, the PMM strategy yields a monthly Sharpe ratio of 0.34 , which outperforms the VM (0.27) and WML (0.08) strategies, respectively. In contrast, the PMM strategy produces moderate performance during periods of non-contraction, represented by a lower monthly Sharpe ratio relative to the VM strategy (0.35 versus 0.39). Analyses show similar outperformance of our PMM strategy from 2000 to 2018 compared to the dynamic WML strategies in Daniel and Moskowitz (2016) from 2000 to 2013, confirming the persistence of risk-adjusted momentum beyond 2013.

Further, we find evidence of a relatively higher correlation between the PMM strategy and the Univariate Trend-Following (UTF) strategy³ compared to the VM strategy. This is consistent with the view that the enhanced performance of the PMM strategy may be attributable to elements of trend following. In addition, we examine the sensitivity of the PMM strategy to conventional factors and economic risks based on the q-factors from Hou et al. (2015) and the economic activity factors from Liu and Zhang (2008) and Maio and Philip (2018). Results reveal that the adjusted R-squared values from our analyses for the PMM strategies are consistently the smallest compared to the VM strategy across all model specifications, which lends strong support that the dynamic exposures to our long and short portfolios lead to a portfolio that is less sensitive to the q-factors. This implies that the PMM strategy may have potential applications in the risk management of momentum-based strategies.

The unconstrained leverage implicit in the PMM strategy makes its outperformance questionable as an investment strategy in practice. As such, we further implement a version of the PMM strategy with 200% leverage constraint (PMMC). Although the PMMC strategy moderately underperforms the PMM strategy (annualized

Sharpe ratio of 1.11 versus 1.23) over the whole sample period, both strategies continue to outperform the VM strategy (annualized Sharpe ratio= 1.02). The implementation of the PMMC strategy is broadly in line with those hedge funds trading in equities and event driven strategies as reported by (Ang et al., 2011) (page 6, 2011). These limitations on leverage exist in practice either through controls imposed through the prime brokerage that the firm is involved in, or the federal regulations governing margin requirements such as The Federal Reserve Board's Regulation T (Reg T).

The remainder of the paper is organized as follows. Section 2 discusses related literature. Section 3 describes the data and tests plain momentum. Section 4 provides the development of our PMM strategy and examines their performance. Section 5 further assesses the outperformance of our partial-moment-scaled momentum against volatility-scaled momentum. Section 6 explains the mechanisms that underpin the risk-adjusted outperformance of our PMM strategy. Section 7 concludes the study.

2. Related literature

The literature has shown that momentum returns are related to partial moments, especially lower partial moments (Baltas and Kosowski, 2012; and Daniel et al., 2012). We employ the downside realized partial moment (RPM^-) and the upside realized partial moment (RPM^+) in the context of momentum trading strategies. We hypothesize that if volatility-based momentum strategies, such as those in Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016), have been shown to counter extreme momentum losses following times of market crashes; as a consequence, further momentum profits and better risk management can be achieved by utilizing the connection between downside realized volatility and future volatility as shown by Barndorff-Nielsen et al. (2010). Other applications of partial moments to finance are included, for example, in Anthonisz (2012) and Bali et al. (2014).

Barndorff-Nielsen et al. (2010) find that future volatility is related more to past negative returns than past positive returns. In doing so, the authors develop a volatility measure called "realized semi-variance," which decomposes the direction of the quadratic variation in asset prices, termed "realized variance." Andersen et al. (2001) and Barndorff-Nielsen and Shephard (2002) define the realized variance as the sum of the squared returns to estimate the quadratic variation in high frequency asset prices. As such, the negative and positive returns of asset prices are used to compute a downside (RS^-) and an upside (RS^+) realized semi-variance, respectively. A review of the semi-variance literature is presented in Sortino and Satchell (2001). Further development of the economic theory underpinning this risk measure is set out in Pedersen and Satchell (2002). These directional volatility measures are found to capture the asymmetrical properties of volatility experienced by asset prices. Hedge funds might employ downside realized semi-variance in the context of risk management. These investors might have short positions in the market, and a drop in price therefore yields a positive return, with the corresponding measure of risk being RS^- . Baruník et al. (2016) extend this idea to construct asymmetric volatility spillover indexes and reveal high levels of asymmetric spillover among the most liquid US stocks in seven sectors. As we argue in Section 3, it is more appropriate that this semi-variance is called partial moments of order 2. Upper and lower partial moments are what has been termed upside and downside realized semi-variance, respectively.

Literature has further shown that momentum interacts with business conditions, market states, market liquidity, investor

³ The UTF strategy is based on longing (shorting) the market when its past 12-month cumulative excess return is positive (negative).

sentiment in addition to market volatility. Chordia and Shivakumar (2002) demonstrate that economic expansionary periods may explain profits in the US equities market, but the literature is inconclusive on whether momentum profits are positive or negative during periods of contraction. This points towards the relevance of partial moment effects on momentum. Cooper et al. (2004) find that momentum profits depend “critically” on the state of the market. They show that a 6-month momentum strategy is profitable only following periods of gains in the US market during the sample period, from 1929 to 1995, consistent with the predictions of overreaction models in Daniel et al. (1998) and Hong and Stein (1999).⁴ Relevant to our work, Min and Kim (2016) find that investors who employ momentum strategies are exposed to greater downside risk, face high levels of expected market risk premium and experience economically large losses during market downturns, supporting the findings in Ang et al. (2001). Thus, we contribute to this literature by considering upside and downside risk as conditioning variables.

3. Data and momentum portfolio construction

The data used in this study are sourced from the center for Research in Security Prices (CRSP) and the Kenneth R. French Data Library. The monthly and daily US equity data for the period January 1926 to December 2018 are sourced from the CRSP. Our sample includes common stocks (CRSP share code 10 or 11) of all firms listed on the NYSE, Amex, and Nasdaq (CRSP exchange code 1, 2 or 3). We use the value-weighted index of all listed firms in the CRSP and the 1-month Treasury bill rate as the proxy for the market portfolio and the risk-free rate, respectively. We obtain the 1-month Treasury bill rate for the same period as that of the US equity data from the Kenneth R. French Data Library.

We follow a method similar to Jegadeesh and Titman's (1993) trading model to construct our WML portfolio with a zero-net position. First, in month t , all valid sample stocks are ranked based on their past 11-month formation period returns from month $t-12$ to $t-2$, and then sorted into ten decile portfolios according to the NYSE breakpoints as each decile portfolio contains an equal number of NYSE firms.⁵ We define a valid sample stock as one which has share price, number of shares outstanding, and a minimum of 8 monthly returns during the 11-month formation period. Subsequently, we buy the best-performing portfolio (winners) and short the worst-performing portfolio (losers) for the 1-month holding period from month $t-1$ to t . The value-weighted holding period logarithmic returns are calculated for computing our momentum profits; this strategy rebalances at the end of every month. There is a 1-month lag between the formation period and the holding period to avoid short-term reversals (e.g., Jegadeesh, 1990; Lehmann, 1990). This allows us to be consistent with Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016).

4. Partial moment momentum strategies

In this section, we provide the development of our PMM strategies and test their performance. Barroso and Santa-

Clara (2015) and Daniel and Moskowitz (2016) present evidence that, by scaling the weights of WML portfolios, this new scaling strategy improves performance. Rather than scaling WML portfolios, we construct the PMM strategy, which can be viewed as an extension of the dynamic volatility-based momentum strategy in Barroso and Santa-Clara (2015), who do not differentiate between upside and downside risk. We extend this class of strategies by tilting our strategy long or short towards favorable/ unfavorable volatility signals and holding an offsetting position in cash.

4.1. Partial moments and realized semi-variance

The proceeding discussion describes a particular asset, say asset k . In what follows we shall suppress the k subscript. We can construct the sample realized volatility using equi-spaced data for $[t-1, t]$, which we define as RV_t where t refers to the end of the period that the asset's return is calculated. This is defined for $n + 1$ prices or their logarithms to define n returns, r_i , $i = 1, \dots, n$. We define RV_t as:

$$RV_t = \sum_{i=1}^n r_{i,t}^2, \quad (1)$$

We call this the realized variance for $[t-1, t]$, which is known to be a consistent estimator of quadratic variation (of asset k) if we assume the prices are generated by a particular class of processes; see Barunik et al. (2016) for more details. The properties of RV_t and related measures can be found in Andersen et al. (2001), Barndorff-Nielsen and Shephard (2002) and Barndorff-Nielsen et al. (2010). Following the literature, we define two statistics, negative semi-variance RS_t^- and positive semi-variance RS_t^+ , as:

$$RS_t^- = \sum_{i=1}^n r_{i,t}^2 I(r_{i,t} < 0), \quad (2)$$

$$RS_t^+ = \sum_{i=1}^n r_{i,t}^2 I(r_{i,t} \geq 0), \quad (3)$$

where $I()$ is the indicator function, these being sample lower and upper partial moments of order 2 with truncation at zero in both cases, $r_{i,t}$ is the return of stock i in day t , and n is the number of valid sample stocks in day t . There is an identity, $RV_t = RS_t^- + RS_t^+$.

If the population has a mean of zero, then we might hope to treat these quantities as estimators of the unconditional population variance and the unconditional lower and upper semi-variances of the process over the interval $[t-1, t]$, which we denote by σ^2 , σ^{2-} , and σ^{2+} , respectively. Hopefully, $\sigma^2 = \sigma^{2-} + \sigma^{2+}$. However, the difficulty here is that, for example, $E(RS_t^+)$ is not equal to σ^{2+} but rather $\sigma^{2+} + E^2[\sum_{i=1}^n r_{i,t} I(r_{i,t} \geq 0)]$. This is because

$\sigma^{2+} = E[\sum_{i=1}^n r_{i,t}^2 I(r_{i,t} \geq 0)] - E^2[\sum_{i=1}^n r_{i,t} I(r_{i,t} \geq 0)]$ with an analogous definition for σ^{2-} . This means that descriptions in terms of partial moments seem more appropriate. Thus, we refer to these two statistics as upper and lower partial moments (RPM_t^+ and RPM_t^-). This follows if we define $E(RV_t) = E[\sum_{i=1}^n r_{i,t}^2]$, $E(RPM_t^-) = E[\sum_{i=1}^n r_{i,t}^2 I(r_{i,t} < 0)]$, and $E(RPM_t^+) = E[\sum_{i=1}^n r_{i,t}^2 I(r_{i,t} \geq 0)]$.

Barroso and Santa-Clara (2015) argue that momentum volatility is strongly forecastable relative to other styles and consequently, WML variance is better forecasted than market variance. We show in Appendix B that when underlying returns are white noise, the forecastability of momentum volatility is connected to underlying kurtosis. Thus, as an alternative to momentum volatility, we might

⁴ Liquidity, analyst coverage, trading volume, and firm credit ratings also affect momentum. See, for instance, Hong et al. (2000), Lee and Swaminathan (2000), Avramov et al. (2007), and Avramov et al. (2016).

⁵ Using NYSE breakpoints rather than all-firm breakpoints can limit the side effects of small and illiquid stocks that are barely traded in practice. This is consistent with Barroso and Santa-Clara (2015) and how the Kenneth R. French Data Library constructs its “10 Portfolios Formed on Momentum.” Our ten decile portfolio returns are almost identical to those of (2-12) momentum returns on the Kenneth R. French Data Library. We repeat all analyses using all-firm breakpoints momentum returns as a robustness check. The results are consistent with those based on our original WML strategy. Our WML returns using NYSE breakpoints are moderately lower than momentum returns on the same basis using all-firm breakpoints.

also use a volatility based on a market index to guide the scaling.⁶ Hence, we define our RPM_t^+ and RPM_t^- as:

$$RPM_t^+ = \frac{21}{126} \sum_{j=1}^{126} r_{d_{t-j}}^2 I(r_{d_{t-j}} \geq 0), \quad (4)$$

$$RPM_t^- = \frac{21}{126} \sum_{j=1}^{126} r_{d_{t-j}}^2 I(r_{d_{t-j}} < 0), \quad (5)$$

where $r_{d_{t-1}}$ is the daily return of the market index in the last trading day of month t and $I(\cdot)$ is the indicator function.

4.2. PMM strategies in detail

Recent studies have enhanced momentum strategies by weighting the momentum positions using volatility in various ways (Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016). These methods do not differentiate between upside or downside risk and typically are, by construction, net zero funds. Practitioner methods based on an optimization approach, however, scale momentum mean forecasts by their standard deviations. We extend this class of strategies by tilting our strategy long or short toward favorable/unfavorable volatility signals and holding an offsetting position in cash.

We define our PMM strategy as a net zero portfolio long $\varphi_1(RPM_t^+, RPM_t^-)$ in winners and short $\varphi_2(RPM_t^+, RPM_t^-)$ in losers with an offsetting position of $\varphi_1(RPM_t^+, RPM_t^-) - \varphi_2(RPM_t^+, RPM_t^-)$ in cash; we call this portfolio p . We denote its return at time $t + 1$ as:

$$r_{p,t+1} = \varphi_1(RPM_t^+, RPM_t^-) r_{w,t+1} - \varphi_2(RPM_t^+, RPM_t^-) r_{l,t+1} + (\varphi_2(RPM_t^+, RPM_t^-) - \varphi_1(RPM_t^+, RPM_t^-)) r_{f,t+1}, \quad (6)$$

where $r_{w,t+1}$, $r_{l,t+1}$, and $r_{f,t+1}$ are the returns at time $t + 1$ to the winners, losers, and “cash” portfolios, respectively.

There is no obvious guidance as to the functional form of $\varphi_1(RPM_t^+, RPM_t^-)$ and $\varphi_2(RPM_t^+, RPM_t^-)$, but we might expect $\varphi_1(RPM_t^+, RPM_t^-)$ to be increasing in its first argument and decreasing in its second argument and $\varphi_2(RPM_t^+, RPM_t^-)$ to have the opposite properties. Besides, the weights of winner and loser portfolios of our PMM strategies, $\varphi_1(RPM_t^+, RPM_t^-)$ and $\varphi_2(RPM_t^+, RPM_t^-)$, avoid look-ahead bias as the upper and lower partial moments are calculated one period before that of the portfolio returns.

Furthermore, we might want to normalize them, as in Barroso and Santa-Clara (2015), in terms of some target volatility, σ_{tar} . We pick the same target annualized volatility of 12% to maintain consistency. With these considerations in mind, we require the constraint to be:

$$\varphi_1(RPM_t^+, RPM_t^-) + \varphi_2(RPM_t^+, RPM_t^-) = \frac{2\sigma_{tar}}{\sqrt{RV_t}}, \quad (7)$$

which is broadly in accord with Barroso and Santa-Clara (2015). We choose the weights $\varphi_1(RPM_t^+, RPM_t^-)$ and $\varphi_2(RPM_t^+, RPM_t^-)$ as follows:

$$\varphi_1(RPM_t^+, RPM_t^-) = \frac{2\sigma_{tar}}{\sqrt{RV_t}} \left(\frac{RPM_t^+}{RPM_t^+ + RPM_t^-} \right), \quad (8)$$

$$\varphi_2(RPM_t^+, RPM_t^-) = \frac{2\sigma_{tar}}{\sqrt{RV_t}} \left(\frac{RPM_t^-}{RPM_t^+ + RPM_t^-} \right), \quad (9)$$

If $RPM_t^+ = RPM_t^-$, then we have a conventional long-short portfolio with scaling $\frac{\sigma_{tar}}{\sqrt{RV_t}}$, which is used in the VM strategy presented

in Barroso and Santa-Clara (2015, p. 115, formula (5)). The computed weights, $\varphi_1(RPM_t^+, RPM_t^-)$ and $\varphi_2(RPM_t^+, RPM_t^-)$, are then employed to create the VM, PMM and PMMC strategies.

Formulae (7) to (9) could be criticized in that we might want to shrink our exposure on the downside when RPM_t^- is large, however, such an approach would be more consistent with the empirical result that momentum profits mainly result from the long side of a portfolio. For instance, Jegadeesh and Titman (1993) state that “the abnormal performance of the zero-cost (momentum) portfolio is due to the buy side of the transaction rather than the sell side” (p. 77). Moskowitz and Grinblatt, (1999) argue that “industry momentum strategies appear to profit mostly on the buy side” (p. 1272). See Chan et al. (1996), Jegadeesh and Titman (2001), and Israel and Moskowitz (2013) for further evidence.

Furthermore, it might be argued that in choosing functions that are homogeneous of degree zero, some of the partial moment information is lost. The argument in favor of increasing one's exposure on the downside is that, in prospect theory, agents are usually deemed risk-loving on the downside. Ang et al. (2006) conclude that “the behavioral framework of Kahneman and Tversky's (1979) loss aversion preferences and the axiomatic approach taken by Gul (1991) disappointment aversion preferences allow agents to place greater weights on losses relative to gains in their utility functions” (p. 1192). The authors also argue that “agents who place greater weight on downside risk demand additional compensation for holding stocks with high sensitivities to downside market movements” (p. 1191). The arguments are further supported by Lettau et al. (2014).

In practice, leverage is an issue in long-short portfolios. Many institutional hedge funds have strict restrictions on leverage, with 200% leverage being a typical upper bound. We justify our PMMC strategy as capturing the actual constraints in leverage that equity hedge funds would experience through their interactions with brokers, and their general inability to borrow in unlimited amounts. The previous popularity of 130–30 funds provides evidence that leverage is not unconstrained in practice.⁷ Leverage is defined as the sum of absolute value of long and short weights so that our abovementioned strategies have a leverage as in Formula (7).

If the leverage as in Formula (7) exceeds the upper bound, to rescale our weights to obey the 200% leverage condition, we need to change our scaling to:

$$\varphi_1(RPM_t^+, RPM_t^-) = 2 \left(\frac{RPM_t^+}{RPM_t^+ + RPM_t^-} \right), \quad (10)$$

$$\varphi_2(RPM_t^+, RPM_t^-) = 2 \left(\frac{RPM_t^-}{RPM_t^+ + RPM_t^-} \right), \quad (11)$$

with corresponding positions in cash.

Fig. 1 plots the 3-month moving average value of the weights of winner and loser portfolios of the PMM and PMMC strategies over the whole sample period. Panel A shows the weights of the PMM strategy. The average (standard deviation) of the weights for the winners and losers, φ_1 and φ_2 , are 1.05 (0.52) and 0.93 (0.38), respectively. In particular, the weight of winners (φ_1) ranges between 0.16 and 3.76, and, on average, has a higher exposure to the winners of plain momentum. Panel B shows the weights of the PMMC strategy which obeys the 200% leverage condition. On average, the exposures of winners and losers in PMMC are reduced in comparison to the PMM and plain momentum strategies. The average (standard deviation) of the weights for the winners and losers are 0.88 (0.31) and 0.79 (0.24), respectively. For both PMM

⁶ We note that when market volatility rather than WML volatility is used to scale WML, the target volatility (σ_{tar}) is less closely attained.

⁷ “130–30” represents a leveraged long-short mutual fund strategy that allows fund managers to hold short positions up to 30% of the initial investment and to use the corresponding funds to take long positions. Other popular strategies of this type include but are not limited to “150–50” and “120–20.”

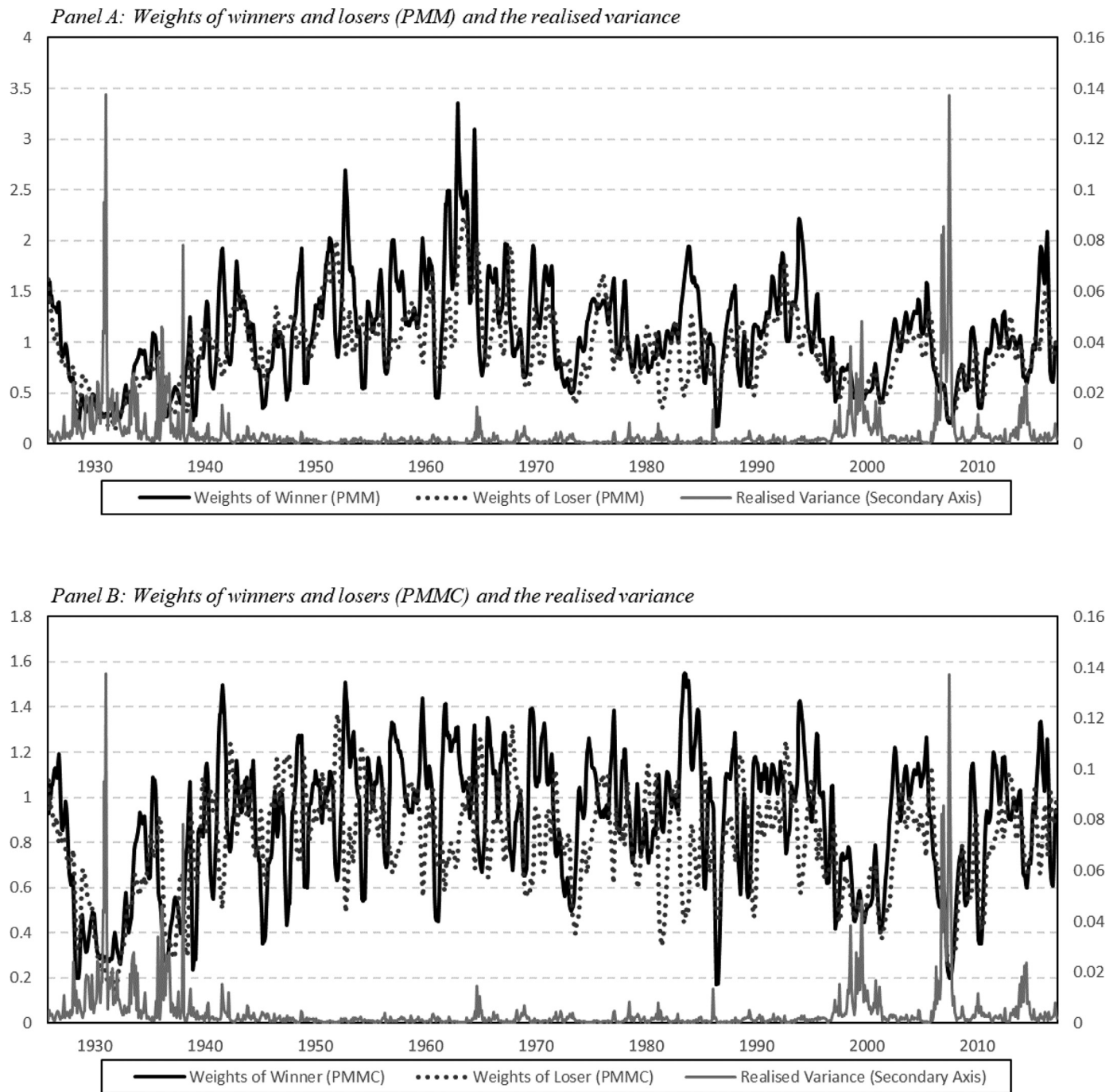


Fig. 1. Weights of Partial Moment Momentum.

We present the 3-month moving average of the weights of the PMM and PMMC strategies in Panels A and B, respectively. The weights of winners and losers (φ_1 and φ_2) and the realized variance of the plain momentum returns are plotted over the whole sample period of 1927–2018, respectively. We tilt both strategies long or short toward favorable/ unfavorable volatility signals measured by upper and lower partial moments as shown in Eq. (8) and (9) for the PMM strategy in Panel A, and Eq. (10) and (11) for the PMMC strategy if leverage exceeds the upper bound in Panel B, respectively.

and PMMC strategies, the weights for the winners and losers are reduced to a relatively low level in the 1930s and 2007–2009, respectively. This shows that our PMM strategies are shielded from volatility exposure when the plain momentum experiences excessive market volatility.

Table 1 provides a summary of the performance of the PMM, PMMC, WML and VM strategies for the whole sample period from July 1927 to December 2018 and the sub-sample period from January 1963 to December 2018. During the whole sample period and the post-1963 period, the 11×1 WML portfolio earns 14.98% and

16.49% annualized returns, respectively. We employ the approach in Cederburg et al. (2020) to test the significance of the differences in Sharpe ratios and find that the Sharpe ratios of both our PMM strategy significantly outperform the benchmark strategies for the whole sample period and the post-1963 period.⁸ Specifically, the PMM strategy reveals outperformance compared to the VM portfolio, evidenced by both an increase in Sharpe ratio from 1.02 to

⁸ In unreported results, we observe that results are robust for the 1965 to 2018 sample period. Results are available upon request.

Table 1

Performance of partial moments-based momentum strategies.

We report performances of the WML, PMM and PMMC strategies for the whole sample period and a sub-period over 1963–2018. We also report the performance of the VM strategy constructed by Barroso and Santa-Clara (2015). All portfolios are on an 11×1 basis. For each strategy, apart from returns and its respective significance, we also report skewness, kurtosis, the annualized Sharpe ratio, adapted Sortino ratio, the information ratio, and the maximum drawdown. The benchmark returns used to compute the information ratios are based on the annualized returns of the WML strategies. Jarque-Bera reports the statistics and p-values (in parentheses; for instance, the reference point when $p = 0.05$ is 5.991.) of the Jarque-Bera normality test in Jarque and Bera (1987), the null hypothesis of which states that the variable is normally distributed. We follow Cederburg et al. (2020) to test the significance of the differences in Sharpe ratios, and we report the p-values from Jobson and Korkie (1981) tests in brackets. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. .

Strategy	Annualized return (%)	Annualized Sharpe Ratio (SR)	SR difference with WML	SR difference with VM	Adapted Sortino Ratio	Information ratio	Maximum Drawdown (%)	Skewness	Kurtosis	Jarque-Bera
<i>Whole Sample Period: July 1927 to December 2018</i>										
WML	14.98 (***)	0.52	–	–	0.16	–	–95.45	–2.33	17.48	10,414.39(<0.001)
VM	18.19 (***)	1.02	0.50 (0.000)	–	0.58	0.80	–34.25	–0.23	2.06	53.12(<0.001)
PMM	20.44 (***)	1.23	0.70 (0.000)	0.20 (0.013)	0.82	0.91	–33.90	0.03	2.72	83.45(<0.001)
PMMC	18.71 (***)	1.11	0.59 (0.000)	0.10 (0.054)	0.73	0.86	–26.11	–0.10	2.63	49.49(<0.001)
<i>Post-1963 Period: January 1963 to December 2018</i>										
WML	16.49 (***)	0.64	–	–	0.13	–	–80.55	–1.40	7.87	882.48(<0.001)
VM	20.09 (***)	1.13	0.49 (0.000)	–	0.70	0.74	–32.33	–0.12	1.57	44.05(<0.001)
PMM	21.32 (***)	1.32	0.74 (0.000)	0.19 (0.019)	1.12	0.86	–34.15	–0.05	2.70	29.23(<0.001)
PMMC	18.48 (***)	1.21	0.57 (0.000)	0.08 (0.071)	1.09	0.82	–27.58	–0.11	2.17	25.21(<0.001)

1.23 (from 1.13 to 1.32) and an increase in skewness from –0.23 to 0.03 (–0.12 to –0.05) over the whole sample (post-1963) period.⁹ Further, the PMM strategy yields a higher information ratio (0.91 versus 0.80) and a lower maximum drawdown (–33.90% versus –34.25%) compared to the VM strategy during the whole sample period.¹⁰ The post-1963 period shows similar outperformance, with the PMM strategy producing a higher information ratio (0.86 versus 0.74) in comparison to the VM strategy.

While we continue to report Sharpe ratios, we also provide a performance measure called the adapted Sortino ratio, more suited to our different volatility regimes as defined in Appendix A. This measure is expected to better capture the “downside-risk-adjusted” performance. Both PMM strategies report higher adapted Sortino ratios compared to the WML and VM strategies. Our PMM strategies might be useful as an alternative tool in the management of momentum crashes following market downturns.

To check the normality of returns, we also conduct the Jarque-Bera normality test. We note that, in almost all cases, the relatively small p-values of the Jarque-Bera normality test suggest rejection of the null hypothesis; that is, strategy returns are not normally distributed. This is consistent with theory where Kwon and Satchell (2018) show that the returns of the cross-sectional momentum strategy are generically non-normally dis-

tributed. However, the Jarque-Bera statistic of the PMM strategies are reduced compared to the WML strategy. For example, the value of the Jarque-Bera statistic for the post-1963 period decreases from 882.48 to 29.23 for the WML and PMM strategies, respectively. This suggests that our partial moments-based momentum strategies help in further managing momentum tail risk.

We further contrast the cumulative performance of VM and PMM strategies and plot the dollar value of investment in Fig. 2. Both VM and PMM strategies outperform the market and WML over a whole sample period of 1927–2018. The VM strategy performs well in the market “calm” period from 1965 until the end of 1990s, however, the PMM strategy presents a smooth upward trend through the whole sample period and in particular, dominates the market “turbulent” period of 2007–2012. This is consistent with our findings in Table 1 that our PMM strategy generates higher risk-adjusted return in all subperiods and have a relative advantage in momentum crashes following market downturn periods.

Further tests reveal that our PMM strategies are robust across strategy construction and multiple time periods.¹¹ While we do not claim that investors could earn these returns in practice, it seems that partial moments-based strategy is a good way to manage momentum risk and an interesting addition to the investment literature. In conclusion, we find that our PMM momentum strategy tilted long or short towards favorable/unfavorable volatility signals is likely to be more efficient and profitable than a constant volatility strategy, especially during bad times.

5. Further analyses

This section first examines the relative performances of the PMM, VM and WML strategies across market states. We then evaluate the performances by introducing turnover and transaction costs. We also compare the dynamic leverage between the PMM

⁹ Similar outperformance against the VM strategy is found for the PMMC strategy, which earns annualized returns of 18.71% significant at the 1% level in conjunction with annualized Sharpe ratios of 1.11 for the whole sample. Although we observe a lower annualized return of 18.48% for the PMMC strategy compared to the VM strategy for the post-1963 period, the Sharpe ratio of the PMMC strategy (1.21) continues to reveal enhanced performance against the VM strategy (1.13). In unreported results, we find underperformance against the VM strategy if we strictly rescale all weights of the PMM strategy in accordance with the 200% leverage condition.

¹⁰ We further test the performance of the PMMC strategy with a 150% leverage constraint given the deleveraging that occurred in most developed financial markets following the GFC. This strategy slightly underperforms PMMC. However, compared to the benchmark strategies (WML and VM), it shows strong outperformance by revealing lower maximum drawdowns and higher risk-adjusted returns during the whole sample period as well as the post-1963 sample period. We also find consistent outperformance in the risk-adjusted performance for different choices of volatility targets.

¹¹ We also follow a 6-month formation and holding period (6×6) strategy as a robustness check. Like the results in Table 1, both the PMM and PMMC strategies outperform the benchmark portfolios.

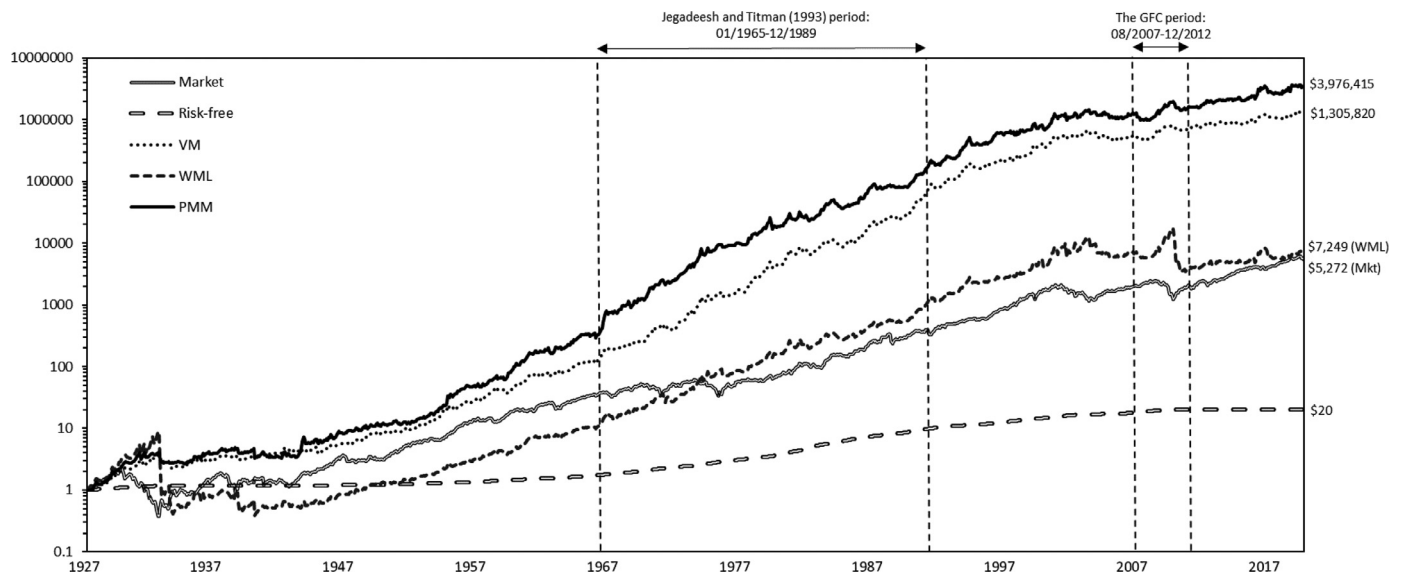


Fig. 2. Cumulative performance over 1927–2018.

In this figure we apply the methodology of Daniel and Moskowitz (2016) to plot the dollar value of investment for five assets including the WML strategy, the VM strategy, the PMM strategy, the market portfolio, and the risk-free asset given a \$1 initial investment in July 1927 over an investment period of 1927–2018. All assets are self-funded and rebalanced monthly. We also assume an investment environment without any transaction costs. In particular, two periods are marked in the figure, the Jegadeesh and Titman (1993) period from January 1965 to December 1989 and the GFC period from August 2007 to December 2012. The starting point of the GFC period of August 2007 represents the onset of financial market turbulence in the equity market and the intervention of the Federal Reserve for the first time since 2001. This is supported by Bekaert, et al. (2014). The end date of the GFC period reflects the recovery of the DJIA, S&P 500, and Nasdaq to their pre-GFC levels.

and VM strategies. Additionally, we test whether the PMM strategy is subject to economic and risk factors in both conditional and unconditional models. Finally, we conduct spanning tests and examine the risk components of the PMM strategy through a decomposition process.

5.1. Performance comparison over market states

Evidence has been shown to support that the PMM strategy may be an improved alternative to the VM strategy to manage downside risk during momentum crashes. We extend the analysis to study the performance of the PMM and PMMC strategies compared to the WML and VM strategies across different market states.

Table 2 presents monthly percentage returns of the strategies stratified by different market states across the whole sample period of 1927–2018 using the following three approaches: (1) by the expected market risk premium (a method similar to Fama and French, 1989; Petkova and Zhang, 2005; Zhang, 2005).¹² Specifically, “Trough (peak)” represents the highest (lowest) 10% periods of the expected market risk premium, respectively. “Recession (expansion)” represents the remaining periods when the expected premium higher (lower) than its average, respectively. “Non-trough” represents the periods when the expected premium is below its 90 percentiles and is complementary to the “Trough” state; (2) we create two additional states to provide a broader coverage of the performance of the strategies during economic good and bad times. The “Bad” state represents the combination of “Trough” and “Recession” periods and the “Good” state represents the combination of “Peak” and “Expansion” periods; and (3) by the U.S. business cycle (a method determined by the National Bureau of Economic Research (NBER) Business Cycle Dating Committee). “Contraction” represents the periods starting at the peak of a busi-

ness cycle and ending at the trough. “Non-contraction” represents the remaining periods.

In Panel A of Table 2, a comparison of the broader definition of economic “Good” and “Bad” times reveals that in “Bad” times, the PMM strategy achieves higher monthly returns (Sharpe ratios) of 1.51% (0.25) at the 1% significance level compared to the WML and VM strategies, which yield monthly returns of 0.18% (0.02) and 1.06% (0.18), respectively. The monthly return of the PMM strategy (1.62%) during economic “Good” times is relatively moderated compared to the WML (2.17%) and VM (1.75%) strategies, respectively at the 1% significance level. Panel B of Table 2 presents momentum performance in accordance with the NBER definition of market cycles, where contraction is the period from the start at the peak of a business cycle and ends at the trough. In this case, the PMM strategy shows outperformance in returns and higher Sharpe ratios compared to both WML and VM during periods of contraction (1.51% monthly return and 0.34 monthly Sharpe ratio for PMM). During periods of non-contraction, the monthly return of PMM (1.53%) is moderately lower than VM (1.59%) and higher than WML (1.23%).¹³ The findings are consistent with the intuition underpinning the mechanism of the PMM strategy, which is geared towards outperformance during momentum crashes through the management of downside risk, and this result comes at the expense of moderated performance during market upturns.

5.2. Turnover and transaction cost

This section evaluates the impact of turnover on our PMM strategies and tests whether they keep outperforming VM strategies after considering transaction costs. We first computed and compared the turnover for WML, VM and our PMM strategies. Portfolio turnover is calculated following the turnover estimation methodology proposed by Barroso and Santa-Clara (2015). By us-

¹² We follow the methodology introduced in Section 2 of Petkova and Zhang (2005) to compute the expected market risk premium by regressing the realized market excess return on conditioning variables.

¹³ The relative performance of PMMC compared to the WML and VM strategies remains robust to the findings of the PMM strategy during economic “Good” and “Bad” times, and during the “Contraction” and “Non-contraction” periods.

Table 2

Momentum profits and market states.

This table reports the performance of the WML, VM, PMM and PMMC strategies with respect to different market conditions. In Panel A, market states are classified by the expected market risk premium following [Petkova and Zhang \(2005\)](#). In particular, “Trough (peak)” represents the highest (lowest) 10% periods of the expected market risk premium, respectively. “Recession (expansion)” represents the remaining periods when the expected premium higher (lower) than its average, respectively. “Non-trough” represents the periods when the expected premium is below its 90 percentiles and is complementary to the “Trough” state. We create two additional states to provide a broader coverage of the performance of the strategies during economic good and bad times. The “Bad” state represents the combination of “Trough” and “Recession” periods and the “Good” state represents the combination of “Peak” and “Expansion” periods. In Panel B, market states are classified by the NBER. “Contraction” represents the periods starting at the peak of a business cycle and ending at the trough. “Non-contraction” represents the remaining periods. Monthly returns in percentage, t-statistics (in parentheses) and monthly Sharpe ratio [in brackets] are reported. The sample periods in all panels are from July 1927 to December 2018. *, **, and *** denote the significance level of 10%, 5%, and 1%, respectively.

	WML	VM	PMM	PMMC
<i>Panel A: Profits across market conditions</i>				
Trough	−3.64 (−2.48)** [−0.24]	−0.52 (−0.73) [−0.07]	−0.63 (−0.83) [−0.08]	−0.80 (−1.09) [−0.10]
Recession	1.13 (3.87)*** [0.18]	1.45 (5.63)*** [0.27]	2.04 (7.79)*** [0.37]	1.64 (7.35)*** [0.32]
Expansion	1.64 (6.61)*** [0.32]	1.73 (7.11)*** [0.34]	1.49 (5.67)*** [0.27]	1.23 (5.68)*** [0.27]
Peak	4.27 (4.97)*** [0.48]	1.82 (3.37)*** [0.32]	2.11 (3.24)*** [0.31]	2.17 (3.70)*** [0.35]
Non-trough	0.83 (3.43)*** [0.11]	1.36 (7.63)*** [0.24]	1.50 (8.03)*** [0.26]	1.21 (7.33)*** [0.23]
Bad	0.18 (0.46) [0.02]	1.06 (4.17)*** [0.18]	1.51 (5.75)*** [0.25]	1.15 (4.91)*** [0.21]
Good	2.17 (8.15)*** [0.35]	1.75 (7.86)*** [0.34]	1.62 (6.53)*** [0.28]	1.42 (6.77)*** [0.29]
<i>Panel B: Profits over periods of contraction and non-contraction</i>				
Contraction	0.95 (1.22) [0.08]	1.24 (2.80)*** [0.27]	1.51 (3.37)*** [0.34]	1.33 (3.13)*** [0.30]
Non-contraction	1.23 (5.60)*** [0.20]	1.59 (8.09)*** [0.39]	1.53 (7.90)*** [0.35]	1.29 (7.67)*** [0.31]

ing such a measure, we are able to compare each leg of the momentum strategies. The turnover of each leg of the PMM portfolio is calculated as follows:

$$\tilde{w}_{i,t-1} = \frac{w_{i,t-1}(1 + r_{i,t})}{\sum_i N_t w_{i,t-1}(1 + r_{i,t})}, \quad (12)$$

$$x_{t,j} = 0.5 \times \sum_i N_t \left| \frac{w_{i,t}}{L_{t,j}} - \frac{\tilde{w}_{i,t-1}}{L_{t-1,j}} \right|, \quad j = \{\text{winner}, \text{loser}\} \quad (13)$$

where $w_{i,t}$, $r_{i,t}$, and N_t are the weight of stock i , the return of stock i , and the number of stocks in the leg $L_{t,j}$ at time t , respectively. The turnover of one strategy is the sum of the turnover of all legs. The winner and loser leg are $L_{t, \text{winner}} = \varphi_1(RPM_t^+, RPM_t^-)$ and $L_{t, \text{loser}} = \varphi_2(RPM_t^+, RPM_t^-)$ respectively.

Results in [Table 3](#) reveal that our PMM strategies report a higher amount of turnover relative to the VM strategy. However, the difference, for instance, at 7.71% for our PMM strategy (82.52%) compared to the VM strategy (74.81%), is not likely to be economically significant.

The above discussion focuses on turnover and it raises the question of transaction costs. Transaction costs can dramatically reduce momentum profits due to the rebalancing of winner and loser portfolios. As argued by [Korajczyk and Sadka \(2004\)](#), notwithstanding that momentum strategies remain profitable after controlling for trading costs, momentum abnormal returns are diluted with market frictions. As shown in the previous section, partial moments-scaled momentum strategies benefit from swift ad-

Table 3

Turnover for momentum-based strategies.

This table reports the monthly turnover for the WML, VM, PMM and PMMC strategies over a sample period of July 1927 to December 2018. We follow the turnover estimation methodology of [Barroso and Santa-Clara \(2015\)](#) which we discussed in [Section 5.2](#). The turnover estimations for each leg of the strategy are presented with the turnover of the strategy, which is the sum of all legs. All figures are in basis points.

	Winner	Loser	Combined
WML	37.79	35.28	73.08
VM	38.30	36.51	74.81
PMM	42.56	39.96	82.52
PMMC	40.38	37.60	77.97

justments to market condition changes due to leverage. As such, whether PMM strategies will remain investable after considering transaction costs is crucial to practical implementation.

We use two proportional transaction cost measures, effective and quoted spreads, to evaluate the impact of transaction cost on the profitability of various momentum strategies. Both spreads are computed using data sourced from CRSP over a sample period of January 1993 to December 2018.¹⁴ The first spread benchmark, effective spread, is measured as the difference between the price

¹⁴ A continuous series of bid and ask data for NYSE (NASDAQ) is available since December 28, 1992 (June 15, 1992). As such, we restrict the starting point of our sample for transaction costs analysis to January 1993.

Table 4

Performance under transaction costs.

This table reports the performance of the WML, VM, PMM and PMMC strategies before and after controlling for transaction costs. Two proportional transaction cost measures, effective and quoted spreads, are utilized. Effective spread is measured as the difference between the price of the trade and the midpoint of quoted bid and ask. Quoted spread is measured as the ratio between quoted bid-ask spread and the midpoint (we use half the quoted spread as the measure for trading cost as the quoted spread is shared by both the buy-side and the sell-side). We record stocks in and out of both the winner and loser portfolios. We also estimate transaction costs on a monthly basis to maintain consistency with our momentum portfolio construction over a sample period of January 1993 to December 2018. The benchmark returns used to compute the information ratios are based on the annualized returns of the WML strategies in Panel A. All reported figures (exclude the t-values) are annualized. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Annualized return (%)	t-value	Sharpe ratio	Adapted Sortino ratio	Information ratio
Panel A: WML Strategy					
Raw return	10.52	1.77(*)	0.35	0.13	–
Net return (effective spread)	7.33	1.45	0.29	0.11	–
Net return (quoted spread)	6.75	1.19	0.24	0.10	–
Panel B: VM Strategy					
Raw return	10.29	3.53(***)	0.69	0.40	–0.15
Net return (effective spread)	7.86	2.70(***)	0.55	0.31	0.25
Net return (quoted spread)	6.91	2.51(**)	0.50	0.29	0.13
Panel C: PMM Strategies					
	Without leverage constraint (PMM)				
Raw return	11.08	3.98(***)	0.79	0.64	0.21
Net return (effective spread)	7.89	2.81(***)	0.60	0.57	0.28
Net return (quoted spread)	6.92	2.70(***)	0.54	0.45	0.15
	With leverage constraint (PMMC)				
Raw return	10.02	3.61(***)	0.67	0.52	–0.19
Net return (effective spread)	7.43	2.53(***)	0.53	0.36	0.08
Net return (quoted spread)	6.60	2.06(***)	0.47	0.31	–0.11

of the trade and the midpoint of quoted bid and ask. The second benchmark, quoted spread, is measured as the ratio between quoted bid-ask spread and the midpoint. Specifically, we use half the quoted spread as the measure for trading cost as the quoted spread is shared by both the buy-side and the sell-side.

Results in Table 4 indicate that the outperformance of our PMM strategies compared to VM strategies remains after controlling for both effective and quoted spreads. All net returns reported by our PMM strategy are higher than the VM strategy and are also significant at 1% level. Compared to the VM strategy, our PMM strategy consistently improves the Sharpe ratio, the adapted Sortino ratio and the information ratio for both raw and net return measures over a sample period of 1993 to 2018. For instance, we observe increases in the Sharpe ratio (from 0.50 to 0.54), the adapted Sortino ratio (from 0.29 to 0.45) and the information ratio (from 0.13 to 0.15) for the PMM strategy compared to the VM strategy for the quoted spread based net return.

5.3. Dynamic leverage

The outperformance raises the question as to why our PMM strategies perform better than the VM strategy. We posit that this is partially because the outperformance benefits from the distinct nature of our strategies whereby the long and short portfolios separately react to a volatility spike and result in a non-zero net exposure.

We first compare the dynamic leverage of VM to that of PMM before, during, and post the Global Financial Crisis (GFC) period in Fig. 3 covering the sample period of January 2003 to December 2012. It is important to note that relative to a conventional long-short structure which one might hope to be market neutral through net weights of zero, our PMM portfolio is designed to have non-zero net weights. The PMM strategy has an average long weight of 0.84 and a short weight of 0.74 relative to the zero-net weight of VM for the 2003–2012 sample period.¹⁵ This flexi-

bility allows our strategy to manage the risk exposure prior periods of market crashes that precede momentum crashes (Daniel and Moskowitz, 2016) by increasing proportionately the short position (through a greater decrease in the long position compared to the short position). Compared to VM, our PMM strategy proportionately decreases the long position more than the short position, which results in an overall increase in the short position prior to momentum crashes after the GFC. Further, this may be explained by its non-neutral net exposure observed during pre-market crash periods. For example, in the low volatility pre-GFC period in early 2007, the PMM strategy displays an earlier sharp reduction in the long weight accompanied by a slower and more moderate reduction in the short leg compared to the timing of the reduction in the weight of the VM strategy. As a robustness check, we also conduct a similar exercise but focus on the momentum crashes which followed the Great Depression period. Results shows similar findings to those in the periods around the GFC.

In Table 5, we systematically compare the dynamic leverage between the PMM, PMMC and VM strategies before momentum crashes and rallies. We present the dynamic leverage of the winner and loser portfolios of both strategies before the 25 biggest monthly gains and losses to the WML portfolio over the 1927–2018 period. It has been suggested that the 2 to 3-year previous market return positively leads the contemporaneous WML profits (Cooper et al., 2004; Daniel and Moskowitz, 2016). As such, we report the dynamic changes in weight in three windows from 36 months prior to the portfolio formation date: (1) from 36 months to 24 months [–36, –24]; (2) from 24 months to 12 months [–24, –12]; and (3) from 12 months to the portfolio formation date [–12, 0]. The average monthly return and the 3-year lagged market return to the 25 biggest monthly gains (losses) to the WML portfolio are 19.63% (–29.06%) and 8.91% (–5.96%), respectively. Results show that there is a distinct relation in the weight of the winner portfolio and the weight of the loser portfolio of our PMM strategies, which makes our overall portfolio negatively exposed to downside risk before momentum crashes. In particular, the PMM strategy reveals that the reduction in weights is dominated by the long-leg in the [–36, –24] and [–24, –12] windows (–13.07% and

¹⁵ We note that our PMM strategies are not characterised by beta-neutrality as the betas for the long and short legs are not equal.

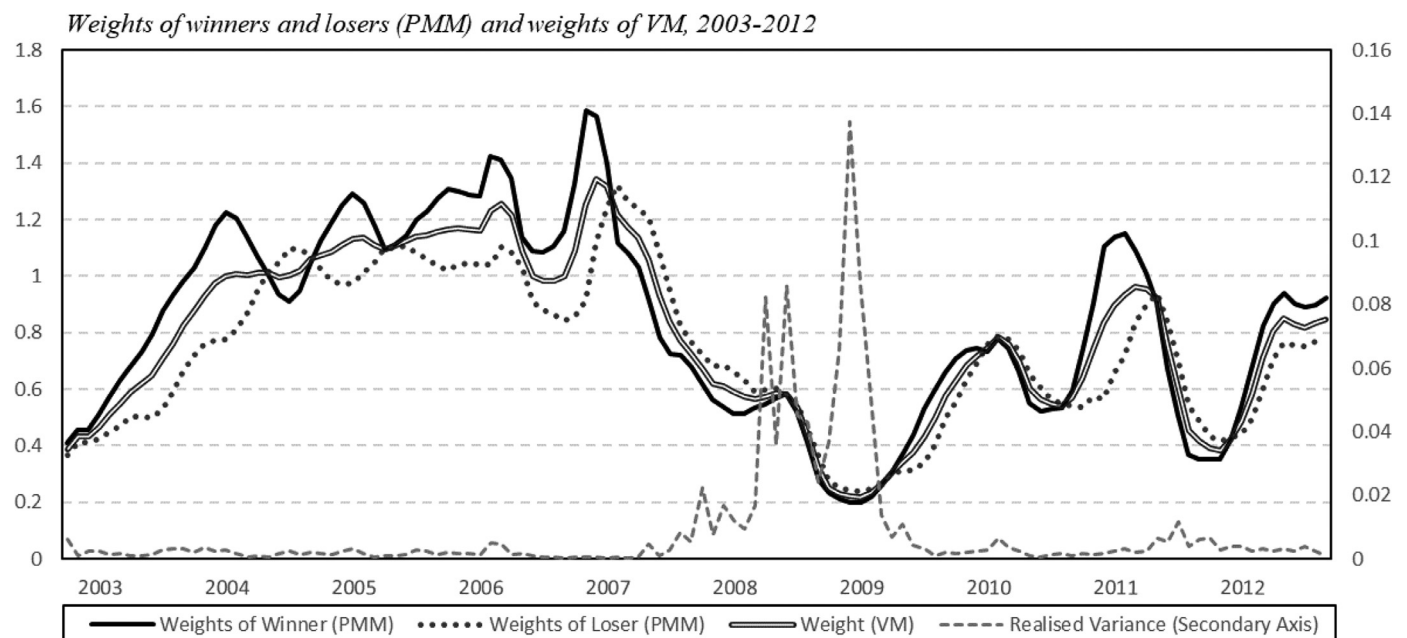


Fig. 3. Comparison of the dynamic leverage between VM and PMM strategies during the GFC.

This figure contrasts the dynamic leverage of our PMM strategy to that of the VM strategy, with a particular focus on the GFC period. We plot the 3-month moving average of the weights of the winner and loser portfolios of our PMM strategy; the weight of the VM strategy; and the realized variance of the plain momentum returns. All series are monthly over a sample period of January 2003 to December 2012.

Table 5

Dynamic leverage before momentum crashes and rallies.

This table presents the comparison of the dynamic leverage between the PMM, PMMC and VM strategies before the 25 biggest monthly gains and losses to the WML portfolio over the 1927–2018 period. We report the average changes in weights in percentage in three windows from 36 months prior to the portfolio formation date. Particularly, for the PMM strategy, the weight of the winner portfolio (φ_1) and that of the loser portfolio (φ_2) follow different specifications in formulae (7) to (8) in Section 4.2; for the PMMC strategy, the weight of the winner portfolio (φ_1) and that of the loser portfolio (φ_2) follow different specifications in formulae (7) to (11) in Section 4.2; for the VM strategy, both the winner and loser portfolios follow the same volatility-scaling regime. All numbers are in percentage.

$\Delta(\text{weight})$	Losses			Gains		
	[−36, −24]	[−24, −12]	[−12, 0]	[−36, −24]	[−24, −12]	[−12, 0]
<i>PMM strategy</i>						
Winner	−13.07	−37.20	−70.27	4.11	19.35	25.74
Loser	5.91	−25.16	−76.85	1.27	12.94	20.73
<i>PMMC strategy</i>						
Winner	−7.23	−25.52	−45.72	3.79	13.09	19.20
Loser	3.64	−18.44	−44.11	0.25	6.08	10.98
<i>VM strategy</i>						
Winner/Loser	−3.57	−31.19	−73.56	2.70	16.15	23.24

−37.20%) compared to the VM strategy (−3.17% and −20.76%), respectively. In the [−12,0] window prior to momentum crashes, the weights of loser portfolio (−76.85%) portfolio of the PMM strategy exhibits a greater percentage reduction compared to the weights of winner portfolio (−70.27%) and the VM strategy (−73.56%). Similar results can be observed before momentum rallies, shown by the long and short legs of our PMM strategy reacting differently in both the [−36, −24] and the [−24, −12] windows. Specifically, our PMM strategy positively reacts to upside market risk before momentum rallies as the weight of the winner portfolio (from 4.11% to 19.35%) increases faster than the weight of the loser portfolio (from 1.27% to 12.94%) in the [−36, −24] and the [−24, −12] windows, respectively.

5.4. Dynamic beta and trend following

As per Daniel and Moskowitz (2016), momentum crashes tend to concentrate following market stress scenarios, due to the condi-

tionally negative beta of WML during market rebounds. In principle a winner-loser portfolio should be beta-neutral except during late stages of the economic cycle, normally the portfolio would report a positive net beta after market rallies due to the higher exposure to the winner portfolio, and a negative net beta after market downturns due to the higher exposure to the loser portfolio. The question is whether the PMM dynamics enable momentum strategy to navigate through the WML crash periods. Further, Goyal and Jegadeesh (2018) debate on the trend versus momentum dynamics and they state that the trend strategy outperforms because it takes a time-varying net long investment. Given that our PMM strategies do not have zero net exposure if we put the offsetting position of cash aside, we examine whether PMM strategies outperform due to their dynamic betas to the underlying market, and thereby benefit from underlying market trends.

Fig. 4 plots the rolling beta estimates of the long-short portfolio of the PMM strategy over the whole sample period and a subsample period during the GFC. The betas are computed us-

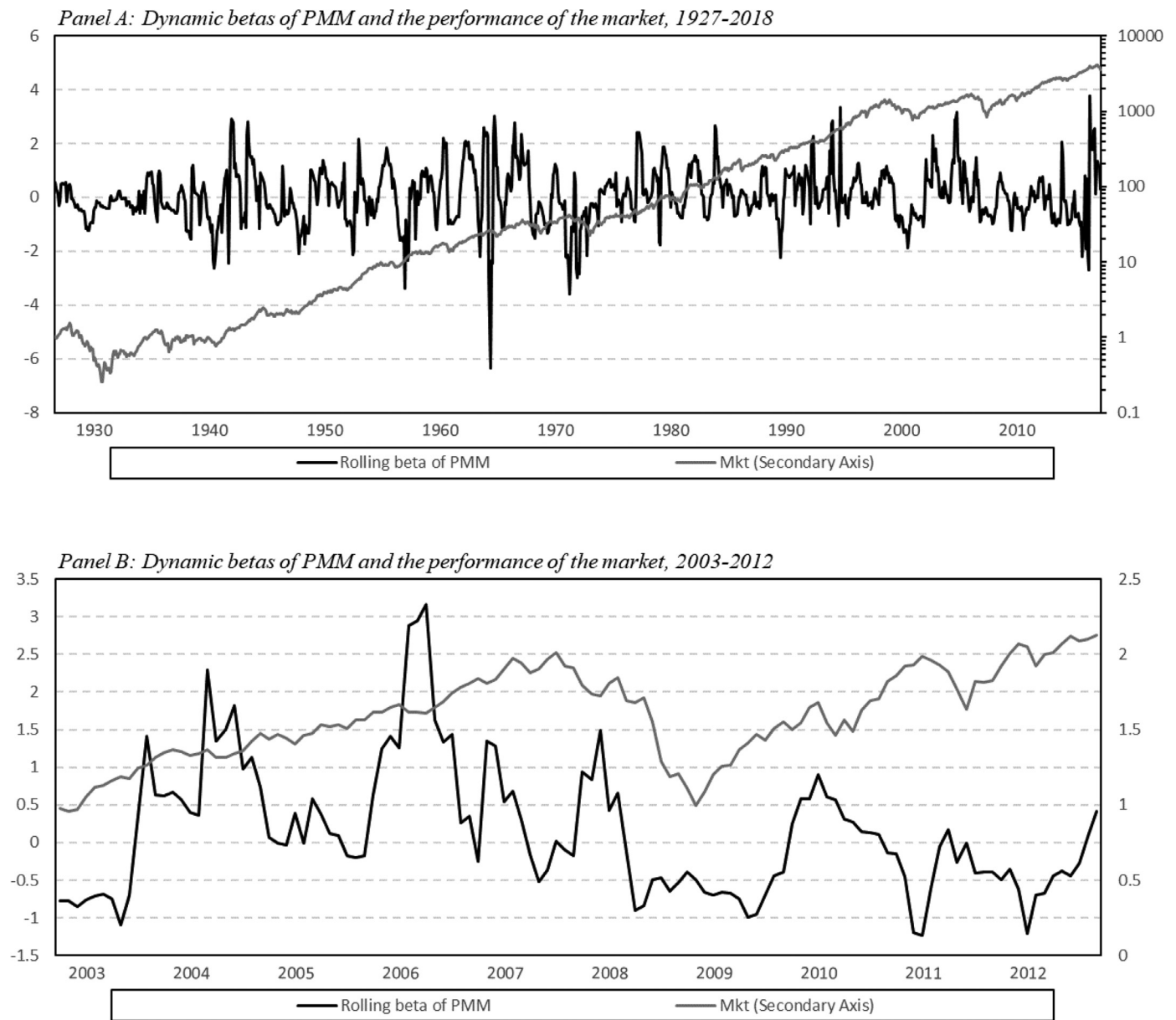


Fig. 4. Rolling betas of the long and short legs.

We plot the beta estimates of the PMM long-short portfolio over the whole sample period of 1927–2018 and a subsample period of 2003–2012 containing the GFC in Panels A and B, respectively. The time-varying betas are estimated following [Daniel and Moskowitz \(2016\)](#) by regressing the 6-month rolling excess returns of PMM long-short portfolio on the excess return of the market. We also plot the dollar value of investment for the market portfolio given a \$1 initial investment in July 1927 (January 2003) over an investment period of 1927–2018 (2003–2012) in Panels A and B, respectively.

ing a 6-month rolling regression model with the value-weighted index of all firms in the CRSP used as the proxy for the market portfolio. Panel A shows that the PMM rolling betas fluctuate across the whole sample period and we observe positive betas during market upturns (increasing market portfolio value) and negative betas during market downturns (decreasing market portfolio value). The positive (negative) betas represent the asymmetrical exposure of the PMM portfolio to the long (short) leg because market upturns are predominantly signified by positive (negative) semi-variance of returns. For instance, Panel B reveals that the rolling betas of PMM long-short portfolios during the turbulent GFC period (late 2008 to mid-2009) are substantially negative. Overall, the PMM strategy exhibits non-zero dynamic betas during market upturns and downturns which allows it to achieve a more

favorable exposure to market trends compared to the WML and VM strategies.

Additionally, [Table 6](#) presents the findings of two correlation tests in relation to the PMM strategies. Panel A shows that the PMM and PMMC strategies have higher correlations with the UTF strategy (0.357 and 0.385) compared to the VM (0.306) strategy. Thus, we might argue that the outperformance of PMM strategies benefits from the wider beta differences between its long and short legs; as well as an element of trend following. As such, our PMM strategies appear to benefit from some of the attractive features of time-series momentum. Panel B reveals that the upper partial moments (RPM^+) of our PMM strategies are positively and significantly associated with positive future market returns (0.301) at the 1% level whilst lower partial moments (RPM^-)

Table 6

Correlation tests

This table reports results of two correlation tests. Panel A presents the correlation coefficients between a UTF strategy and four momentum-based strategies (WML, VM, PMM and PMMC). The UTF strategy is constructed by longing the market when its past 12-month cumulative excess return is positive and by shorting the market when its past 12-month cumulative excess return is negative. The strategy is rebalanced monthly and we use the value-weighted index of all firms in the CRSP as the proxy for the market portfolio. Panel B reports the correlation coefficients between realised volatility, upper/lower partial moments and positive/negative future market returns. We compute the positive (negative) future market return as the cumulative return of all positive (negative) daily market returns in the following month. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample period covers 1927–2018.

Panel A:Correlations between the UTF, WML, VM, PMM and PMMC strategies					
	UTF	WML	VM	PMM	PMMC
UTF	1.000				
WML	0.457 (***)	1.000			
VM	0.306 (***)	0.851 (***)	1.000		
PMM	0.357 (***)	0.788 (***)	0.929 (***)	1.000	
PMMC	0.385 (***)	0.835 (***)	0.911 (***)	0.978 (***)	1.000
Panel B:Correlations between partial moments and future market returns					
	RV	RPM ⁺	RPM [−]	Mkt_future ⁺	Mkt_future [−]
RV	1.000				
RPM ⁺	0.969 (***)	1.000			
RPM [−]	0.980 (***)	0.900 (***)	1.000		
Mkt_future ⁺	0.192 (**)	0.301 (***)	−0.101 (*)	1.000	
Mkt_future [−]	−0.036	−0.087	0.265 (**)	0.027	1.000

are positively and significantly associated with negative future market returns (0.265) at the 5% level. In comparison, the combined realized volatility is found to be positively and significantly correlated with positive market returns (0.192) at the 5% level. These findings highlight the capacity of our PMM strategies to distinguish between upside and downside risks to enhance risk management.

5.5. Exposure of momentum profits to factor risk and lagged market returns

The following regression specifications investigate the effects of economic and risk factors on the profits of the WML, VM, PMM and PMMC strategies:

$$ret_{i,t} = \alpha_i + \beta_{1,i}MKT_t + \beta_{2,i}ME_t + \beta_{3,i}I/A_t + \beta_{4,i}ROE_t + \varepsilon_{i,t}, \quad (14)$$

$$ret_{i,t} = \alpha_i + \beta_{1,i}MKT_t + \beta_{2,i}ECONACTIVITY_t + \varepsilon_{i,t}, \quad (15)$$

with $i = WML, VM, PMM, PMMC$. where $ret_{i,t}$ represents the profits of strategy i (WML, VM, PMM, PMMC) at time t ; MKT_t represents the Fama-French market factor at time t ; ME_t , I/A_t , and ROE_t represent the size, value-growth, investment, and profitability q-factors of Hou et al. (2015) at time t ; specifically, MKT_t represents the market factor proxied by the market excess return at time t ; ME_t represents the size factor proxied by the return difference between small and large stocks at time t ; I/A_t represents the investment factor proxied by the return difference between low investment and high investment stocks at time t ; and ROE_t represents the profitability factor proxied by the return difference between high profitability and low profitability stocks at time t ; $ECONACTIVITY_t$ represents the economic activity factors at time t , which is proxied by industrial production growth (IP) and change in capacity utilization (CU), respectively. The data on q-factors and economic activity factors are obtained from the q-factors data library at global-q.org and the Federal Reserve Bank of St. Louis (FRED), respectively.

Table 7 presents the exposure of momentum profits of the WML, VM, PMM and PMMC strategies to the q-factors from Hou et al. (2015) in specification (2) and the economic activity factor from Liu and Zhang (2008) and Maio and Philip (2018) in specifications (3) and (4). We observe that the adjusted R-squared values for the PMM and PMMC strategies are consistently the smallest compared to the WML and VM strategies across all model specifications, which suggests that the q-factors are less correlated with the PMM and PMMC strategies in comparison to the WML and VM strategies. Our PMM and PMMC strategies yield coefficients for IP (0.125 and 0.113) and CU (0.103 and 0.097) that are both significant at least at the 5% level in specifications (3) and (4), respectively. This suggests that an economic activity factor may likely explain momentum profits for our PMM strategies. With the exception of specification (4) for our PMM strategy, the MKT loading is not significant in specifications (2), (3) and (4) for our PMM strategies. The finding implies that the PMM strategies are likely to be characterized by a low exposure to the market which makes our PMM strategies “pure-alpha” and their profitability free from market conditions. The ROE factor is shown to appropriately explain momentum anomalies, as supported by the significant coefficients at the 1% level for all four strategies.

We explore the exposure of momentum profits of the WML, VM, PMM and PMMC strategies to Fama-French factors and extend the analysis to lagged cumulative 3-year market returns using the following regression specifications:

$$ret_{i,t} = \alpha_i + \beta_{1,i}LAGMKT_t + \beta_{2,i}MKT_t + \beta_{3,i}SMB_t + \beta_{4,i}HML_t + \varepsilon_{i,t}, \quad (16)$$

$$ret_{i,t} = \alpha_i + \beta_{1,i}LAGMKT_t^+ + \beta_{2,i}LAGMKT_t^- + \beta_{3,i}MKT_t + \beta_{4,i}SMB_t + \beta_{5,i}HML_t + \varepsilon_{i,t}, \quad (17)$$

with $i = WML, VM, PMM, PMMC$. where $ret_{i,t}$ represents the profits of strategy i (WML, VM, PMM, PMMC) at time t ; $LAGMKT_t$ represents the 3-year lag market cumulative return at time t ; $LAGMKT_t^+$ ($LAGMKT_t^-$) represents the 3-year lag market cumulative positive (negative) return at time t , with a value equal to $LAGMKT_t$ if $LAGMKT_t$ is positive (negative), or 0 otherwise;¹⁶ MKT_t , SMB_t and HML_t represent the Fama-French market, size, and value-growth factors at time t , respectively.¹⁷

We observe from Table 8 that the coefficients of $LAGMKT$, $LAGMKT^+$ and $LAGMKT^-$ are not highly significant (at 5% level or above) under the PMM and PMMC strategies for all four model specifications. WML reveals significance in the coefficients of $LAGMKT$ related variables under all four specifications, which is consistent with the findings of Cooper et al. (2004) that the profitability of plain momentum is subject to lagged market returns. The VM strategy shows that only the $LAGMKT^+$ coefficient is not highly significant (at 5% level or above) under specifications (2) and (4). For instance, in specification (4), the adjusted R-squared for the PMM strategy is 6.00% compared to 27.51% and 10.13% for WML and VM, respectively. Further, specifications (3) and (4) of PMM reveal that the coefficients of the market (MKT) and size (SMB) factors are less significant compared to WML and VM,¹⁸ suggesting that PMM is relatively less exposed to these two risk factors. Overall, momentum profits under PMM are less exposed to lagged market returns compared to WML and VM, which imply that the performance of the PMM strategy is likely to be more agnostic to lagged market upturns and downturns. The VM strategy shows improvement over the WML in terms of exposure to lagged positive market returns ($LAGMKT^+$ not significant in (2) and (4)) but it shows a highly significant level of exposure to lagged negative market returns (0.033 and 0.029 at 1% and 5% significance levels, respectively).

5.6. Return spreads, conditional multifactor models and spanning tests

We conduct tests of the risk-adjusted performance of the high-minus-low return spreads of the WML, VM, PMM and PMMC strategies by computing the alphas of associated unconditional and conditional multifactor models. The regressions for the scaled models are as follows:

¹⁶ Specifically, in each month t , $LAGMKT_t = (PMKT_{t-1} - PMKT_{t-37})/PMKT_{t-37}$, where $PMKT_t$ represent the price of the value-weighted index of all listed firms in the CRSP in month t . Besides, $LAGMKT_t^+ = LAGMKT_t I(LAGMKT_t > 0)$ and $LAGMKT_t^- = LAGMKT_t I(LAGMKT_t < 0)$.

¹⁷ We do not provide interpretations of the intercept values because they do not represent pricing errors. The reason is that at least one of the factors in the models in Equations (15) to (17) is not traded, that is, it does not represent a monthly excess stock return. This includes $ECONACTIVITY$, as well as $LAGMKT$, $LAGMKT^+$, and $LAGMKT^-$. The coefficients obtained from the time-series regressions associated with the macroeconomic models only enable us to estimate factor loadings, which are then employed in second stage regressions to yield pricing error estimates. For instance, see Maio and Santa-Clara (2017) and Cooper, Ma, and Maio (2021). We thank a referee for this comment.

¹⁸ The PMMC strategy exhibit moderately stronger findings, where the coefficients of all risk factors are consistently insignificant, with exception of the HML factor.

Table 7

Exposure of momentum profits to factors

This table reports the regression results of the WML, VM, PMM and PMMC strategies on risk factors. *MKT* is the Fama-French market factor. *ME*, *I/A*, and *ROE* are risk factors from Hou et al. (2015). Specifically, the market factor (*MKT*) is proxied by the market excess return; the size factor (*ME*) is proxied by the return difference between small stocks and large stocks; the investment factor (*I/A*) is proxied by the return difference between low investment stocks and high investment stocks; the profitability factor (*ROE*) is proxied by the return difference between high profitability stocks and low profitability stocks. *IP* and *CU* are economic activity factors from Liu and Zhang (2008) and Maio and Philip (2018), in which *IP* represents the growth rate of industrial production whilst *CU* represents the change in capacity utilization. Estimated coefficients, t-statistics (in parentheses), and the adjusted R-squared values are reported. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample period is from January 1967 to December 2018.

	WML				VM				PMM				PMMC			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
<i>Intercept</i>	0.014 (4.98)***	0.004 (1.80)	0.011 (3.36)**	0.001 (3.19)**	0.015 (6.85)***	0.010 (4.25)***	0.012 (5.69)***	0.011 (5.52)***	0.016 (6.57)***	0.010 (3.98)***	0.012 (6.93)***	0.012 (6.87)***	0.013 (6.34)***	0.010 (3.55)***	0.012 (6.36)***	0.013 (6.51)***
<i>MKT</i>	−0.27 (−4.34)***	−0.172 (−2.82)**	−0.293 (−4.35)***	−0.311 (−4.56)***	−0.099 (−2.01)**	−0.031 (−0.62)	−0.083 (−1.82)*	−0.089 (−1.92)*	−0.036 (−0.68)	0.019 (0.35)	−0.068 (−1.57)	−0.071 (−1.80)*	−0.054 (−1.13)	0.005 (0.11)	−0.061 (−1.07)	−0.063 (−1.10)
<i>ME</i>		0.318 (3.72)***				0.181 (2.56)**				0.261 (3.42)***				0.248 (3.65)***		
<i>I/A</i>		−0.047 (−0.34)				−0.040 (−0.35)				−0.048 (−0.38)				0.005 (0.04)		
<i>ROE</i>		1.472 (14.30)***				0.956 (11.23)***				0.982 (10.73)***				0.925 (11.31)***		
<i>IP</i>			0.129 (2.11)**				0.107 (2.03)**				0.125 (2.10)**				0.113 (2.15)**	
<i>CU</i>				0.079 (1.79)*				0.116 (2.08)**				0.103 (1.99)**				0.097 (2.31)***
<i>Adjusted-R²</i>	2.78%	26.68%	3.35%	4.55%	0.49%	17.07%	2.78%	2.19%	0.00%	15.35%	2.15%	1.76%	0.00%	16.82%	1.15%	1.03%

Table 8

Exposure of momentum profits to 3-year lag market cumulative return.

This table reports the regression results of the WML, VM, PMM and PMMC strategies on lagged market returns and risk factors. *LAGMKT* reports the 3-year lag market cumulative return. *LAGMKT+* (*LAGMKT-*) with a value equal to *LAGMKT* if *LAGMKT* is positive (negative), or 0 otherwise. *Mkt*, *SML* and *HML* are Fama-French risk factors. Estimated coefficients, t-statistics (in parentheses), and the adjusted R-squares are reported. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample period is from July 1927 to December 2018.

	WML				VM				PMM				PMMC			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
<i>Intercept</i>	0.005 (2.13)**	0.014 (3.01)***	0.01 (3.63)***	0.016 (3.90)***	0.013 (7.44)***	0.015 (8.03)***	0.014 (8.56)***	0.018 (8.79)***	0.014 (11.06)***	0.015 (11.76)***	0.015 (10.33)***	0.017 (12.07)***	0.013 (14.75)***	0.014 (13.02)***	0.016 (13.85)***	0.016 (13.10)***
<i>LAGMKT</i>	0.022 (3.75)***		0.02 (3.46)***		0.015 (2.75)***		0.007 (1.99)*		0.009 (1.76)*		0.006 (1.83)*		0.007 (1.80)*		0.005 (1.72)*	
<i>LAGMKT+</i>		0.017 (2.02)**		0.014 (1.81)*		0.011 (1.70)*		0.010 (1.69)*		0.003 (1.03)		0.003 (0.92)		0.003 (1.00)		0.003 (0.95)
<i>LAGMKT-</i>		0.075 (2.97)***		0.062 (2.76)***		0.033 (2.89)***		0.029 (2.43)**		0.006 (1.27)		0.008 (1.50)		0.007 (1.42)		0.007 (1.39)
<i>MKT</i>	-0.613 (-15.73)***	-0.621 (-16.16)***	-0.697 (-12.19)***	-0.683 (-5.32)***	-0.147 (-5.28)***	-0.153 (-3.54)***	-0.126 (-3.47)***	-0.133 (-1.97)*	-0.032 (-2.25)**	-0.031 (-2.30)**	-0.056	-0.056 (-0.79)	-0.007 (-1.49)	-0.007 (-1.35)	-0.019	-0.019
<i>SMB</i>			-0.173 (-2.04)**	-0.163 (-1.89)**			-0.123 (-2.59)***	-0.133 (-2.72)***			-0.031 (-1.09)	0.037 (-1.15)			0.003 (0.69)	0.003 (0.54)
<i>HML</i>			-0.537 (-10.54)***	-0.563 (-10.55)***			-0.495 (-7.13)***	-0.503 (-8.09)***			-0.307 (-6.70)***	-0.303 (-6.82)***			-0.115 (-6.48)***	-0.117 (-6.52)***
<i>Adjusted-R²</i>	15.83%	16.45%	27.12%	27.51%	4.07%	4.46%	9.10%	10.13%	2.19%	2.31%	5.87%	6.00%	1.48%	1.19%	3.21%	2.86%

$$\begin{aligned}
ret_{i,t} = & \alpha_i + \beta_{1,i}MKT_t + \beta_{2,i}MKT_tVS_{t-1} + \beta_{3,i}MKT_tTB_{t-1} \\
& + \beta_{4,i}MKT_tRD_{t-1} + \beta_{5,i}MKT_tIK_{t-1} + \beta_{6,i}ME_t + \beta_{7,i}ME_tVS_{t-1} \\
& + \beta_{8,i}ME_tTB_{t-1} + \beta_{9,i}ME_tRD_{t-1} + \beta_{10,i}ME_tIK_{t-1} + \beta_{11,i}I/A_t \\
& + \beta_{12,i}I/A_tVS_{t-1} + \beta_{13,i}I/A_tTB_{t-1} + \beta_{14,i}I/A_tRD_{t-1} \\
& + \beta_{15,i}I/A_tIK_{t-1} + \beta_{16,i}ROE_t + \beta_{17,i}ROE_tVS_{t-1} \\
& + \beta_{18,i}ROE_tTB_{t-1} + \beta_{19,i}ROE_tRD_{t-1} + \beta_{20,i}ROE_tIK_{t-1} + \varepsilon_{i,t}
\end{aligned} \quad (18)$$

$$\begin{aligned}
ret_{i,t} = & \alpha_i + \beta_{1,i}MKT_t + \beta_{2,i}MKT_tVS_{t-1} + \beta_{3,i}MKT_tRREL_{t-1} \\
& + \beta_{4,i}MKT_tRD_{t-1} + \beta_{5,i}MKT_tNTIS_{t-1} + \beta_{6,i}SMB_t \\
& + \beta_{7,i}SMB_tVS_{t-1} + \beta_{8,i}SMB_tRREL_{t-1} + \beta_{9,i}SMB_tRD_{t-1} \\
& + \beta_{10,i}SMB_tNTIS_{t-1} + \beta_{11,i}HML_t + \beta_{12,i}HML_tVS_{t-1} \\
& + \beta_{13,i}HML_tRREL_{t-1} + \beta_{14,i}HML_tRD_{t-1} + \beta_{15,i}HML_tNTIS_{t-1} \\
& + \beta_{16,i}RMW_t + \beta_{17,i}RMW_tVS_{t-1} + \beta_{18,i}RMW_tRREL_{t-1} \\
& + \beta_{19,i}RMW_tRD_{t-1} + \beta_{20,i}RMW_tNTIS_{t-1} + \beta_{21,i}CMA_t \\
& + \beta_{22,i}CMA_tVS_{t-1} + \beta_{23,i}CMA_tRREL_{t-1} + \beta_{24,i}CMA_tRD_{t-1} \\
& + \beta_{25,i}CMA_tNTIS_{t-1} + \varepsilon_{i,t}
\end{aligned} \quad (19)$$

with $i = WML, VM, PMM, PMMC$. where $ret_{i,t}$ represents the profits of strategy i (WML, VM, PMM, PMMC) at time t ; MKT_t represents the Fama-French market factor at time t ; ME_t , I/A_t , and ROE_t represent the size, value-growth, investment, and profitability q-factors of Hou et al. (2015) at time t ; SMB_t , HML_t , RMW_t , and CMA_t represent the size, value-growth, investment, and profitability factors in Fama and French (2015, 2016) at time t , respectively; The conditional models incorporate additional lagged variables from Cooper and Maio (2019).¹⁹ Specifically, specification (18) employs the value spread (VS), 1-month T-bill rate (TB), investment-capital ratio (IK), and stock return dispersion (RD) as the lagged variables for the conditional HXZ models; specification (19) utilizes the relative T-bill rate (RREL), net equity expansion (NTIS), VS, and RD as the lagged variables for the conditional FF models.

Table 9 Panel A shows that the PMM strategy produces higher alphas than the VM strategy for all specifications where the coefficients for both strategies are significant at the 5% level or above. For instance, our PMM strategy presents a higher significantly positive alpha of 1.05 compared to the alpha of 0.89 for the VM strategy for the specification based on the conditional HXZ model using "All" four lagged variables. The PMM strategy in Panel B demonstrates higher and significantly positive alphas compared to the VM strategy across all specifications. Generally, minimal variation in the significance of the alphas is shown between each of the two baseline unconditional multifactor HXZ and FF models and their corresponding models that incorporate "All" conditional information.

We follow Daniel and Moskowitz (2016) to conduct spanning tests by running daily regressions of the returns of the PMM and VM portfolios with respect to the WML portfolio and the FF factors.²⁰ The unconditional and conditional regressions for the PMM strategy relative to the WML strategy are as follows:

$$\begin{aligned}
PMM_t = & \alpha + \beta_1MKT_t + \beta_2SMB_t + \beta_3HML_t + \beta_4RMW_t \\
& + \beta_5CMA_t + \beta_6WML_t + \varepsilon_t,
\end{aligned} \quad (20)$$

¹⁹ Note that we did not use the conditional CAPM as one of our chosen conditional models because "conditional CAPM is not a valid answer for explaining cross-sectional equity risk premia" (Lewellen and Nagel, 2006; Cooper and Maio, 2019).

²⁰ We thank an anonymous referee for this suggestion. We choose not to utilize the HXZ factors because: (1) the ROE factor spans the baseline momentum (WML) factor (Hou et al., 2019); and (2) the ROE factor strongly interacts with VM, PMM, and PMMC returns as shown in Table 7.

$$\begin{aligned}
PMM_t = & \alpha + \beta_1MKT_t + \beta_2MKT_tMS_{t-1} \\
& + \beta_3SMB_t + \beta_4SMB_tMS_{t-1} + \beta_5HML_t + \beta_6HML_tMS_{t-1} \\
& + \beta_7RMW_t + \beta_8RMW_tMS_{t-1} + \beta_9CMA_t + \beta_{10}CMA_tMS_{t-1} \\
& + \beta_{11}WML_t + \beta_{12}WML_tMS_{t-1} + \varepsilon_t,
\end{aligned} \quad (21)$$

and similarly for the PMM strategy relative to the VM and PMMC strategies. where MKT , SMB , HML , RMW , and CMA represent the market, size, value-growth, investment, and profitability factors in Fama and French (2015, 2016), respectively; MS represents the bear market indicator with a value which equals 1 if the cumulative past two-year market return is negative or 0 otherwise. The regressions for the VM and the PMMC strategies are conducted in a similar way.

In Table 10, each factor is interacted with a market stress indicator in specifications (2), (4), (6) and (8) to produce their conditional betas. Table 10 Panel A demonstrates significantly positive alphas for the spanning tests of our PMM portfolio with respect to the WML (alpha=16.92% and 13.81%) and VM (alpha=4.11% and 2.76%) portfolios at the 1% (specifications (1) and (2)) and 5% (specifications (3) and (4)) levels, respectively. In contrast, Panel C reveals that the spanning tests of the returns of the VM portfolio with respect to the FF+PMM model (alpha=1.61%) and the conditional version of that model (alpha=1.59%) in specifications (5) and (6) are weakly significant at the 10% level. Overall, the results of the spanning tests show that the outperformance of our PMM strategy is unlikely to be driven by dynamic exposure to the WML or VM strategies, or to the FF factors.²¹

6. Outperformance of the pmm strategy

This section discusses the possible mechanisms that underpin the outperformance of the PMM strategy. The risk-scaling component of the PMM and VM strategies reduces the exposure of their long-short portfolios in periods of high volatility. The general outperformance of PMM and VM over the WML strategy that is demonstrated in Table 1 relies on the negative relation between volatility and future returns, which is particularly true during times of high volatility when markets fall more than when they rise. In these downturn periods, volatility scaling acts to reduce total portfolio volatility but not returns. As such, the PMM and VM strategies yield relatively higher Sharpe ratios compared to the WML strategy.

Moreover, the economic consequences of volatility in falling or rising markets are not equal. The PMM strategy takes advantage of this characteristic by managing its exposure to the market and allows for unequal scaling of positions between the long-side and short-side weights according to the relative levels of positive and negative semi-variances, which subsequently produces further outperformance compared to the VM strategy, particularly during market downturns. A comparison of the dynamic leverage between the PMM and VM strategies shows that, in the low volatility period preceding the (high volatility) period of a market crash, the PMM strategy is characterized by an accelerated reduction in the long leg with a slower and milder reduction in the short leg. Consequently, during periods of market downturns (high volatility) when negative semi-variance outweighs positive semi-variance, portfolio scaling is tilted towards the short-side weight relative to the long-side weight (such as the negative rolling betas during the GFC period shown in Panel B of Fig. 4). This flexibility in the scaling mechanism distinguishes the PMM strategy from the VM strategy, which is restricted by equal positions on long-side and short-side weights.

²¹ Similar findings in Table 10 Panel B demonstrates that our PMMC strategy is unlikely to be driven by dynamic exposure to the WML strategy, or to the FF factors.

Table 9

Return spreads with unconditional and conditional factor models

This table reports the alphas (intercepts, in percentages) for the high-minus-low return spreads of the WML, VM, PMM and PMMC strategies using unconditional and conditional factor models. We employ the four-factor model (HXZ) from [Hou et al. \(2015\)](#) and the five-factor model (FF) from [Fama and French \(2015, 2016\)](#) as the unconditional baseline models. We follow [Cooper and Maio \(2019\)](#) to choose the lagged variables for the conditional models. Specifically, for the conditional HXZ models reported in Panel A, value spread (VS), 1-month T-bill rate (TB), investment-capital ratio (IK), and stock return dispersion (RD) are utilized as the lagged variables; for the conditional FF models reported in Panel B, relative T-bill rate (RREL), net equity expansion (NTIS), VS, and RD are utilized as the lagged variables. "All" presents the conditional models using all four respective lagged variables. Significance t-statistics are indicated in parentheses below the alphas. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample period is from January 1967 to December 2018.

Panel A: Alphas from the unconditional HXZ four-factor model and its conditional specifications						
Strategy	HXZ	VS	TB	RD	IK	All
WML	0.35 (1.23)	0.03 (0.02)	0.33 (0.98)	0.19 (1.17)	0.07 (0.89)	0.06 (0.91)
VM	0.97 (3.05)***	0.72 (1.99)**	0.45 (1.24)	0.37 (0.74)	0.21 (0.08)	0.89 (4.59)***
PMM	1.01 (4.10)***	0.65 (1.92)*	0.52 (1.77)*	0.43 (0.93)	0.75 (2.01)**	1.05 (4.65)***
PMMC	0.96 (4.02)***	0.80 (2.37)**	0.56 (1.80)*	0.39 (0.91)	0.79 (2.18)**	0.93 (4.33)***
Panel B: Alphas from the unconditional FF five-factor model and its conditional specifications						
Strategy	FF	RREL	NTIS	RD	VS	All
WML	1.02 (2.20)**	1.63 (2.38)**	1.29 (2.15)**	1.33 (3.97)***	0.98 (2.45)**	1.03 (2.41)**
VM	1.27 (4.79)***	1.35 (2.20)**	1.47 (2.07)**	1.39 (3.90)***	1.02 (2.51)**	1.31 (4.67)***
PMM	1.30 (5.03)***	1.39 (2.13)**	1.50 (1.99)**	1.63 (4.24)***	1.21 (3.85)***	1.35 (4.88)***
PMMC	1.26 (4.66)***	1.32 (2.18)**	1.45 (2.03)**	1.66 (4.25)***	1.22 (3.83)***	1.32 (4.59)***

Table 10

Spanning tests

This table reports the annualized alphas (in percentages) of spanning tests for the VM, PMM and PMMC portfolios with respect to the momentum portfolios and the five-factor model (FF) of [Fama and French \(2015, 2016\)](#). We follow [Daniel and Moskowitz \(2016\)](#) to conduct the spanning tests by running daily regressions of the returns of the PMM, PMMC and VM portfolios on these factors. Each factor is interacted with a market stress indicator to estimate the conditional betas. The sample period is from January 1967 to December 2018. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Coefficient	Factor set							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	FF+WML	FF+WML conditional	FF+VM	FF+VM conditional	FF+PMM	FF+PMM conditional	FF+PMMC	FF+PMMC conditional
<i>Panel A: Dependent variable = returns to the PMM strategy</i>								
Alpha	16.92	13.81	4.11	2.76			1.43	1.60
t(Alpha)	(9.27)***	(7.09)***	(2.30)**	(1.99)**			(2.10)**	(2.43)**
<i>Panel B: Dependent variable = returns to the PMMC strategy</i>								
Alpha	9.48	8.05	2.05	1.49	−0.35	−0.33		
t(Alpha)	(4.50)***	(4.13)***	(1.94)*	(1.50)	(−0.60)	(−0.58)		
<i>Panel C: Dependent variable = returns to the VM strategy</i>								
Alpha	12.29	9.77			1.61	1.59	1.97	1.84
t(Alpha)	(6.13)***	(4.90)***			(1.70)*	(1.72)*	(2.05)**	(1.91)*

The effectiveness of the flexible scaling scheme of the PMM strategy is further highlighted by the significantly positive alphas observed in the spanning tests of the PMM strategy in [Table 10](#), which implies that the outperformance is unlikely to be a result of the dynamic exposure to the WML portfolio, the VM portfolios, or the FF factors.

Further, the correlation results in Panel B of [Table 6](#) suggest that lower (upper) partial moments are associated with negative (positive) future market returns. Additionally, the combined realized volatility is only weakly associated with positive market returns and is not significantly correlated with negative market returns. These results suggest that the outperformance of our PMM

strategy may be attributable to its ability to differentiate between upside and downside risks.

Finally, we decompose the realized variance of our PMM strategies into systematic and specific components.²² The outcome shows that, on average, the systematic components only account for 13.75% and 11.60% of the total risk for the PMM and PMMC strategies, respectively, and the specific components of the risk ac-

²² We follow the methodology of [Barroso and Santa-Clara \(2015\)](#) and use a market model to decompose the realized variance of our PMM strategies ($RV_{i,t}$) into systematic ($RV_{RMRF,t}$) and specific ($\sigma_{\varepsilon,i,t}^2$) components as shown: $RV_{i,t} = \beta_{i,t}^2 RV_{RMRF,t} + \sigma_{\varepsilon,i,t}^2$, with $i = PMM, PMMC$.

count for most of the outperformance. This finding is consistent with the results in Table 7, which suggests that the performance of the PMM strategy is free from market conditions.

7. Conclusion

We have demonstrated good performance for cross-sectional momentum strategies, using information in past partial moments. In particular, we construct a strategy based on partial moment scaling whereby we load up on long (short) positions if upper (lower) partial moments are forecasted to be relatively large.

Our approach, relative to what a fund manager might do, is conservative in the sense of using minimal calculations. While we use CRSP data, which involve many stocks too small or illiquid to be considered by a fund of reasonable size, we do not use any out-of-sample data in our results to avoid look-ahead bias. Our volatility estimation is based on daily data from the past six months.

Our work has some similarities to Daniel and Moskowitz (2016) in that we use volatility targets and time varying volatility estimates. However, we use upper and lower partial moments as opposed to parametric volatility models such as GARCH. The advantage of our approach is that it may reduce estimation error in that it focuses on semi-parametric volatility. We differentiate between upside and downside risks which have different investment implications. Finally, our PMM strategy retains the benefit of a net-zero exposure.

Our results show that our PMM strategy, which essentially move our long-short positions depending upon the relative magnitude of upside and downside volatilities, outperform benchmark WML and VM strategies particularly during periods of financial turbulences. Evidence from dynamic leverage patterns during the pre-GFC low volatility period reveals that the PMM strategy, compared to the VM strategy, is characterized by a proportionate increase in the short position. Subsequently, negative rolling betas during the high volatility GFC period demonstrates that the flexible portfolio scaling mechanism of the PMM strategy tilts towards the short leg relative to the long leg.

Second, our PMM strategy possesses a higher correlation with the UTF strategy compared to the VM strategy, which suggests that the outperformance of the PMM strategy may be partially explained by elements of trend following. Evidence suggests that the PMM strategy is less exposed to factor risks compared with WML and VM strategies. It is possible that the dynamics that are captured by style factors are essentially purged by the changing weights that we use in our portfolio construction. Furthermore, the PMM strategies, like the VM strategy, reduce tail risk in periods of high volatility, as opposed to other risk-managed strategies such as BAB.

Overall, we have derived strategies that both complement and extend the recent literature that uses volatility measures to enhance cross-sectional momentum. Extensions of our analysis include applications of our procedures to other momentum strategies, including time-series momentum and relative strength strategies. Further applications such as application to other asset classes can also be investigated. We leave open the question of whether partial moments are fully or partially priced by markets. Linking partial moments to expected utility has been examined in the literature (for instance, see Fishburn, 1977).

CRedit authorship contribution statement

Yang Gao: Conceptualization, Methodology, Software, Data curation. **Henry Leung:** Investigation, Writing – original draft, Supervision. **Stephen Satchell:** Writing – review & editing, Validation.

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Appendix A. Information ratio, Sharpe ratio and adapted Sortino ratio

We use the Sharpe ratio (Sharpe, 1994), where we define an appropriate formulation for a long-short portfolio as:

$$\text{Sharpe ratio} = \frac{\mu_L - \mu_S}{sd(LS)}, \quad (\text{A.1})$$

where μ_L and μ_S are returns of a long and short portfolio, respectively. $sd(LS)$ is the standard deviation of the long-short portfolio.

The information ratio is calculated using sample moments. In similar spirit to the Sharpe ratio, we replace the riskless rate of return by benchmark returns, which may be plain momentum returns, depending upon context, as follows:

$$\text{Information ratio} = \frac{\mu_A - \mu_B}{sd(r_A - r_B)} \quad (\text{A.2})$$

This calculation involves computing the mean sample returns of the fund of interest, A , likewise the benchmark, B , and the standard deviation of their relative returns.

We define the adapted Sortino ratio as:

$$\text{Adapted Sortino ratio} = \frac{\text{Excess Return}}{2 * \text{Downside SemiDeviation}}, \quad (\text{A.3})$$

where $\text{Excess Return} = \sum_{i=1}^N \text{Excess Return}_i / N$, and

$$\text{Excess Return}_i = R_i - \text{Desired Target Return}_i, \quad (\text{A.4})$$

Sortino and Price (1994), for example, define sample downside semi-deviation as²³:

$$\text{Downside SemiDeviation} = \sqrt{\frac{\sum_{i=1}^N (\text{Excess Return}_i - \bar{\text{Excess Return}})^2}{N} I(\text{Excess Return}_i < 0)}, \quad (\text{A.5})$$

where Excess Return is the portfolio's excess return on the $\text{Desired Target Return}$; $I()$ is the indicator function, this being an empirical excess return lower than zero. Our adaptation differs from the standard Sortino ratio which has a target return equal to the riskless rate. This has the attractive property that if downside and upside standard deviations are equal, we would recover the Sharpe ratio. Thus, we place our version of the Sortino ratio into a broadly similar scale to the Sharpe ratio.

²³ See, in particular Sortino and Price (1994, p. 61, third paragraph). Sortino and Price (1994) refer to this "downside risk" measure as downside deviation, but we consider downside semi-deviation to be more appropriate.

Appendix B. Momentum forecastability and two alternative approaches

On one hand, Barroso and Santa-Clara (2015) suggest that momentum volatility has strong power in forecasting momentum returns by using 126 days overlapping momentum returns to estimate volatilities. In our Sections 4 and 5, we show consistently that using momentum partial moments to risk manage WML produces even stronger outperformance.

However, on the other hand, Barroso and Santa-Clara (2015) also indicate that “there are alternative methods to manage the risk of momentum and exploit its predictability” as in their robustness checks section. There, the authors state that “we find that forecasting the risk of momentum with the realized variance of the market in the previous six months produces very similar results”. Hence, using other volatility measures to scale WML is possible. There is also some theoretical support for the notion that momentum returns are volatility dependent (see, for example, He et al., 2016). One might argue that part of the forecastability is artificial. To observe this, one could oversimplify and assume that a momentum strategy can be described as a position in an asset where the “weight” of the asset is equal to the last period’s return, which is coined as a relative strength portfolio by Lo and Mackinlay (1990). This can be identified as a version of single variable time-series momentum. Thus, R_t , the return of the strategy, is equal to $r_t r_{t-1}$.

Under the assumption that r_t is iid (μ, σ^2) , it follows by elementary calculation that:

$$\text{Var}(R_t) = E(r_t^2 r_{t-1}^2) - E^2(r_t r_{t-1}) \quad (\text{B.1})$$

$$\text{Cov}(R_t, R_{t-1}) = E(r_t r_{t-1}^2 r_{t-2}) - E(r_t r_{t-1}) E(r_{t-1} r_{t-2}) \quad (\text{B.2})$$

This leads to an autocorrelation coefficient as $\rho = \frac{\mu^2 \sigma^2}{\sigma^4 + 2\mu^2 \sigma^2}$. If we interpret the signal-to-noise ratio of the strategy as $SN = \frac{\mu}{\sigma}$, then $\rho = \frac{SN^2}{1 + 2SN^2}$. Thus, if the strategy has a signal-to-noise ratio of 0.5, then the returns will appear to have an autocorrelation coefficient of 0.17 although the underlying data are pure white noise.

Furthermore, as SN becomes large, we reach an upper bound for ρ of $\frac{1}{2}$. Assuming that $E(r_t) = 0$, now it turns to:

$$\text{Var}(R_t^2) = E(r_t^4 r_{t-1}^4) - E^2(r_t^2 r_{t-1}^2) \quad (\text{B.3})$$

$$\text{Cov}(R_t^2, R_{t-1}^2) = E(r_t^4 r_{t-1}^4 r_{t-2}^2) - E((r_t^2 r_{t-1}^2) E((r_{t-2}^2 r_{t-1}^2))) \quad (\text{B.4})$$

$$\text{Let } \mu_j = E(r_t^j), \text{ then } \rho = \frac{\frac{1}{\mu_j^2}}{\frac{1}{\mu_j^2} + 1}.$$

This shows that when underlying returns are white noise, the lower the forecastability of momentum volatility, the higher the kurtosis. If one takes this to be the kurtosis of semi-annual index returns, one might expect a number near 5, and the (spurious) autocorrelation might thus be of a similar magnitude to that seen before.

The above discussions might suggest that momentum is forecastable even when returns are independently and identically distributed (iid). We observe that, when underlying returns are white noise, the forecastability of momentum volatility is connected to underlying kurtosis, and the higher the kurtosis, the lower the forecastability. Thus, as an alternative to momentum volatility, we might also use a volatility based on market index to guide the scaling.

In addition to using partial moments of the market portfolio as the volatility (partial moment) measure of our PMM strategies, we consider using partial moments of the plain momentum from Barroso and Santa-Clara (2015) and the out-of-sample forecasts of the inverse of the volatility of the WML return spread

from Barroso et al. (2020) as the scaling instrument.²⁴ In unreported results, we find both robustness tests support the findings of Barroso and Santa-Clara (2015) and Barroso et al. (2020), indicating that WML volatility and out-of-sample forecasts of the inverse of the volatility of the plain momentum may be alternative measures to manage momentum risks.

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²⁴ We thank an anonymous referee for this suggestion. The details of the implementation of the strategy can be found in Barroso et al. (2020).

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